

Lab: [Optional] Setting Up Cassandra Docker



Estimated time needed: 15 minutes

Objectives

After completing this lab, you will be able to:

- Pull the Cassandra image to your local machine
- Validate the Cassandra database using the Command Line Interface (CLI)

Running the lab

This instructional lab does not require a Skills Network lab environment. You will follow the steps in the lab on your local machine.

Pull the DockerHub image

Run this docker command to pull the Cassandra image and run it on port 9042

```
docker run --name cassandra -d -p 9042:9042 cassandra:5.0
```

Run the docker ps command to confirm that the container has started.

```
docker ps
```

You should see an output similar to the following:

```
theia@theiadower-captainfedo1:/home/project/cassandra-songs$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                    NAMES
1a5233447169   docker.io/library/cassandra:5.0    cassandra -f            13 seconds ago Up 13 seconds  0.0.0.0:9042->9042/tcp  cassandra
```

In future docker commands, you will use the name `Cassandra` when referring to the running container.

Bash into the container

Exec into the docker container using the exec command

```
docker exec -it cassandra bash
```

Validate CLI

From the Cassandra running container, you will use the following command to validate that `cqlsh` is available in this docker container.

```
cqlsh --version
```

You should see a version number similar to this output:

```
root@1a5233447169:/# cqlsh -version
cqlsh 6.2.0
```

The output of the command shows that the **cqlsh** shell is available.

Execute Cassandra command

Next, use the shell to describe all the keyspaces currently in the database. You can pass the command `describe keyspaces;` directly to the `cqlsh` shell.

```
root@1a5233447169:/# cqlsh -e 'describe keyspaces;'
WARNING: cqlsh was built against 5.0-beta1, but this server is 5.0. All features may not work!
system      system_distributed system_traces system_virtual_schema
system_auth  system_schema      system_views
```

Congratulations! You have set up the Cassandra 5.0 database and validated that you can execute **cqlsh** shell commands. You will now upload embedded documents into a new table.

Authors

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